

LECTURE FOUR

Bayesian Inference for Simple Linear Regression

$$(x_1, y_1), \dots, (x_n, y_n)$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$$y_i \stackrel{\text{iid}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

$$\text{likelihood: } \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left[-\frac{1}{2\sigma^2} (y_i - \beta_0 - \beta_1 x_i)^2\right]$$

$$\propto \sigma^{-n} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right]$$

$$= \sigma^{-n} \exp\left[-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right]$$

$$\text{priors: } \beta_0, \beta_1, \log \sigma \stackrel{\text{iid}}{\sim} \text{Unif}(-C, C)$$

(think of $C = \infty$)

$$f(\beta_0, \beta_1, \sigma) = \frac{\mathbb{I}\{-C < \beta_0 < C\}}{2C} \frac{\mathbb{I}\{-C < \beta_1 < C\}}{2C} \frac{\mathbb{I}\{e^{-C} < \sigma < e^C\}}{2C\sigma}$$

$$\log \sigma \sim \text{Unif}(-C, C) \Rightarrow \sigma \sim$$

$$\propto \frac{\mathbb{I}\{-C < \beta_0, \beta_1, \log \sigma < C\}}{\sigma}$$

$$\text{posterior } \propto \text{prior} \times \text{likelihood}(\beta_0, \beta_1, \sigma)$$

$$\propto \frac{\mathbb{I}\{-C < \beta_0, \beta_1, \log \sigma < C\}}{\sigma} \left(\frac{1}{\sigma}\right)^n \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right)$$

$$= I\{-C < \beta_0, \beta_1 < C\} \left(\frac{1}{\sigma}\right)^{n+1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right)$$

Integrate this to get marginal densities of the variables separately.

β_0, β_1 : main parameters

σ : nuisance parameters

$$f(\beta_0, \beta_1)_{\beta_0, \beta_1 | \text{data}} = \int_{-\infty}^{\infty} f(\beta_0, \beta_1, \sigma)_{\beta_0, \beta_1, \sigma | \text{data}} d\sigma \rightarrow \text{LAW OF TOTAL PROBABILITY}$$

$$\propto I\{-C < \beta_0, \beta_1 < C\} \int_{-\infty}^{\infty} \left(\frac{1}{\sigma}\right)^{n+1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) d\sigma$$

$$\approx I\{-C < \beta_0, \beta_1 < C\} \int_0^{\infty} \left(\frac{1}{\sigma}\right)^{n+1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) d\sigma$$

$$\beta = \frac{\sigma}{\sqrt{S(\beta_0, \beta_1)}}$$

$$= I\{-C < \beta_0, \beta_1 < C\} \int_0^{\infty} \left(\frac{1}{\sigma}\right)^{n+1} \left(S(\beta_0, \beta_1)\right)^{-\frac{n+1}{2}} \exp\left[-\frac{1}{2\beta^2}\right] d\beta$$

$$= I\{-C < \beta_0, \beta_1 < C\} \left(S(\beta_0, \beta_1)\right)^{-\frac{n}{2}} \int_0^{\infty} \left(\frac{1}{\beta}\right)^{n+1} \exp\left(-\frac{1}{2\beta^2}\right) d\beta$$

$$\propto I\{-C < \beta_0, \beta_1 < C\} \left[\frac{1}{S(\beta_0, \beta_1)}\right]^{\frac{n}{2}}$$

$$f(\beta_0, \beta_1) \propto \mathbb{I}\{-C < \beta_0, \beta_1 < C\} \left[\frac{1}{S(\beta_0, \beta_1)} \right]^{\frac{n}{2}}$$

$\beta_0, \beta_1 \mid \text{data}$

Posterior mode: Least squares estimators: $\hat{\beta}_0, \hat{\beta}_1$

$$\propto \mathbb{I}\{-C < \beta_0, \beta_1 < C\} \left[\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)} \right]^{\frac{n}{2}}$$

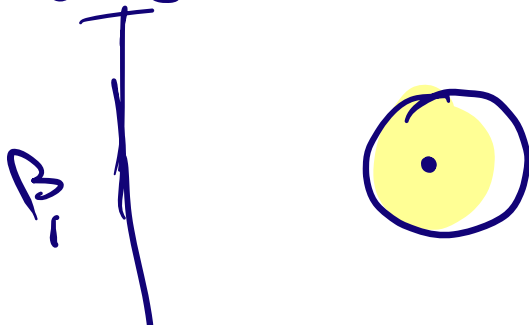
Claim: Posterior is concentrated around the least squares estimator $\hat{\beta}_0, \hat{\beta}_1$.

① If $S(\beta_0, \beta_1) = (1.1) S(\hat{\beta}_0, \hat{\beta}_1)$,
then posterior at (β_0, β_1) : $\left(\frac{1}{1.1} \right)^{\frac{n}{2}}$

② If $S(\beta_0, \beta_1) = (1.01) S(\hat{\beta}_0, \hat{\beta}_1)$
posterior: $\left(\frac{1}{1.01} \right)^{\frac{n}{2}}$

Conclude: Bayesian inference is also based on with $\beta_0, \beta_1, \text{ by } \sigma \sim \text{Unif}(C, C)$

the least squares estimators $\hat{\beta}_0, \hat{\beta}_1$. Posterior typically is highly concentrated: $\left[\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)} \right]^{\frac{n}{2}}$



$$\text{Posterior: } \propto \frac{S(\hat{\beta}_0, \hat{\beta}_1)^{\frac{n}{2}}}{S(\beta_0, \beta_1)} \quad \left. \vphantom{\frac{S(\hat{\beta}_0, \hat{\beta}_1)^{\frac{n}{2}}}{S(\beta_0, \beta_1)}} \right\} \boxed{t\text{-density}}$$

Multiple Linear Regression

y
response

$x_1, x_2, \dots, x_m$
covariates

Data: $(y_i, x_{i1}, x_{i2}, \dots, x_{im}), i=1, \dots, n$

Tabular Data

	y	x_1	$\dots$	x_m
i				

Model: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_m x_{im} + \varepsilon_i$
 $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

(a) Functions of Time as Covariates:

$x_1 = \text{time} \rightarrow x_{i1} = i$
 $x_2 = (\text{time})^2 \rightarrow x_{i2} = i^2$

$$x_3 = \cos(2\pi \times \text{time}) \rightarrow x_{\varepsilon 3} = \cos(2\pi \frac{t}{12})$$

⑥ AutoRegression:

$x_1 = y$ at the previous time $\rightarrow x_{\varepsilon 1} = y_{i-1}$

$x_2 = y$ two time points before $\rightarrow x_{\varepsilon 2} = y_{i-2}$

$$S(\beta_0, \dots, \beta_m) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{\varepsilon 1} - \dots - \beta_m x_{\varepsilon m})^2$$

$\rightarrow$ minimize to get $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m$ (least squares estimator)

$\beta_0, \beta_1, \dots, \beta_m, \log \sigma \stackrel{iid}{\sim} \text{Unif}(-C, C)$

posterior of $\beta_0, \dots, \beta_m \propto \left[\frac{S(\hat{\beta}_0, \dots, \hat{\beta}_m)}{S(\beta_0, \dots, \beta_m)} \right]^{\frac{1}{2}}$

$p = m+1$

Fact: This posterior is a multivariate t-density

Multivariate t-Density

$$\propto \left[\frac{1}{1 + \frac{1}{\nu} (x - \mu)^T \Sigma^{-1} (x - \mu)} \right]^{\frac{\nu + p}{2}}$$

$(x_1, \dots, x_p)$

p : dimension

ν : degrees of freedom

μ : location, Σ : scale or Covariance

$$\left[\left(\frac{S(\hat{\beta}_0, \dots, \hat{\beta}_m)}{S(\beta_0, \dots, \beta_m)} \right)^{\frac{n}{2}} \right]$$

Matrix Notation for Regression

Data: $y_i, x_{i1}, \dots, x_{im} \quad i=1, \dots, n$

$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1}$

↑
vector of response values

$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{pmatrix}$

↓
 $n \times (m+1)$

Values of covariate x_1

`sm.OLS(y, X).fit()`

$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}_{(m+1) \times 1}$

$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_m \end{pmatrix}_{(m+1) \times 1}$

$$S(\beta_0, \dots, \beta_m) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2$$

$$S(\beta) = \| y - \underbrace{X \beta}_{\substack{n \times (m+1) \\ n \times 1}} \|_{(m+1) \times 1}^2$$

$$S(\beta) = \| y - X\beta \|^2$$

$$S(\beta) = (y - X\beta)^T (y - X\beta)$$

$$= y^T y - 2\beta^T X^T y + \beta^T X^T X \beta$$

Minimize to get a formula for least squares:

$$\nabla S(\beta) = \begin{pmatrix} \frac{\partial}{\partial \beta_0} S(\beta) \\ \frac{\partial}{\partial \beta_1} S(\beta) \\ \vdots \\ \frac{\partial}{\partial \beta_m} S(\beta) \end{pmatrix} = -2 \nabla (\beta^T X^T y) + \nabla (\beta^T X^T X \beta)$$

$$= -2 X^T y + 2 X^T X \beta$$

$$= 0$$

$$X^T X \beta = X^T y$$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$$

Facts about $S(\beta)$

$$\textcircled{1} \hat{\beta} = (X^T X)^{-1} X^T y$$

$$\hat{\beta}_1 = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\textcircled{2} S(\beta) = \|y - X\beta\|^2$$

$$= \|y - X\hat{\beta} + X\hat{\beta} - X\beta\|^2$$

$$= (y - X\hat{\beta} + X\hat{\beta} - X\beta)^T (y - X\hat{\beta} + X\hat{\beta} - X\beta)$$

$$= (y - X\hat{\beta})^T (y - X\hat{\beta}) + (\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta)$$

$$= \underbrace{(y - X\hat{\beta})^T (y - X\hat{\beta})}_{S(\hat{\beta})} + \underbrace{(\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta)}_{\text{Cross product term}}$$

Exercise: = 0

$$S(\beta) = S(\hat{\beta}) + (\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta)$$

Posterior β : $\left[\frac{S(\hat{\beta})}{S(\beta)} \right]^{\frac{n}{2}}$

$$= \left\{ \frac{S(\hat{\beta})}{S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta})} \right\}^{\frac{n}{2}}$$

t-density : $\left\{ \frac{1}{1 + \frac{1}{2} (\alpha - \mu)^T \Sigma^{-1} (\alpha - \mu)} \right\}^{\frac{n+p}{2}}$

$$\left\{ \frac{1}{1 + (\beta - \hat{\beta})^T \frac{X^T X}{S(\hat{\beta})} (\beta - \hat{\beta})} \right\}^{\frac{n}{2}}$$

$$p = m+1, \quad \nu + p = n \Rightarrow \nu = n - p = n - m - 1$$

$$\mu = \hat{\beta} : \quad \frac{1}{2} \Sigma^{-1} = \frac{X^T X}{S(\hat{\beta})}$$

$$\Rightarrow \Sigma = \frac{S(\hat{\beta})}{2} (X^T X)^{-1}$$

$$\Sigma = \frac{S(\hat{\beta})}{n - m - 1} (X^T X)^{-1}$$

Posterior of $\beta_0, \dots, \beta_m$: $t_{m+1} \left(\hat{\beta}, \frac{S(\hat{\beta})}{n-m-1} (X^T X)^{-1}, n-m-1 \right)$
 least squares

Simulate $\beta^{(1)}, \beta^{(2)}, \dots, \beta^{(N)}$

$S(\hat{\beta})$: Residual Sum of Squares 2

$$\sum \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_m x_{im} \right)$$

$y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_m x_{im}$: i^{th} residual